

S2CHANGE: Desenvolvimento de mapas de perdas recentes de floresta e mato em Portugal derivados de imagens de satélite

Contrato N.º 3044 de cooperação entre a Direção Geral do Território e o Instituto Superior de Agronomia

Entregável 4.3. Relatório de validação dos mapas nacionais.

(O relatório é redigido em Inglês por conveniência)

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Mapping Cuts and Fires from Sentinel-2 pairs of images

ISA_DGT_INCD_DL_pipeline is the computation pipeline developed at the INCD HPC cluster, operated by LIP, to process complete time series of Sentinel-2 imagery for Portugal.

The goal of ISA_DGT_INCD_DL_pipeline is to produce near-real-time national maps of vegetation loss for areas of forest, bushes or shrubs. ISA_DGT_INCD_DL_pipeline includes a deep learning classifier at the chip level, which consists of 256 by 256 pixels corresponding to an area on the ground of 2560 m by 2560 m. Each chip processing corresponds to a date D, a "before D/after D" pair of temporal composites of 10 band Sentinel-2 images with 10 m resolution.

Polygons of detected vegetation loss for each chip are combined to create a National Vegetation Loss map, where each polygon is labelled with the type of change (Cuts or Fires) and the date.

The following deep learning models were used at the “chip” level:

1. A Swin-YNet architecture, comprising a “Swin Transformer encoder”, more specifically the pre-trained model for Sentinel-2 and fires designated BACDM.
2. A simpler architecture, of the convolutional neural network type with U-net architecture, more specifically an EfficientNet_b2 pre-trained with ImageNet data. The advantage of this second architecture is that it is more light-weight and can run in CPU, since at this point the INCD HPC cluster does not provide GPU nodes.

Since the validation results are very similar, Efficientnet_b2 was adopted as the deep learning component of the ISA_DGT_INCD_DL_pipeline.

Inputs and outputs

The model input for a given date is the pair of 10 bands, 10 m resolution, Sentinel-2 temporal composites on a 256 by 256 pixel “chip”.

The model output is a segmented image with labels “No change” (background) , “Cuts”, and “Fires” and date of detection. It can also be “NoData” when Sentinel-2 imagery is not available at the location and for a significant time span around the date of analysis.

Reference data for independent validation

A lot of effort of the project was centered in acquiring a representative and sound reference data set. There are two main products:

1. A harmonized, vector based, reference data base for Cuts and Burned areas. The burned area part is simply provided by the ICNF official burned area maps (“Territórios ardidos” from GeoCatálogo: https://geocatalogo.icnf.pt/catalogo_tema5.html). The Cuts part is from a combination of sources. All sources of reference data are described at https://github.com/S2change/vegetation_loss/tree/main/data_info/reference_data/Harmonized.

Specifically, the individual data sets that were compiled and harmonized are:

- a) NVG: based on NVG information, completed by semi-automatic verification of dates from a CCD analysis of Sentinel-2 time series, and visually confirmed (see https://github.com/S2change/vegetation_loss/tree/main/data_info/reference_data/NVG)
- b) ICNF: official burned area maps from ICNF
- c) BDR-TNE-300: DGT polygon data base for 300 circular areas in tile T29TNE

d) BDR-TNE-300-Expanded: expansion of a subset of the above for 1 km by 1km polygons

The harmonized data set (including all individual data sets) contains around 11'000 features, ranging from 2017 to 2023. Each feature has an associated time uncertainty interval [data0, data1], where, with high confidence, vegetation loss occurred. Changes were confirmed visually and guided by a CCD (continuous change detection – see https://github.com/S2change/vegetation_loss/tree/main/scripts/pyccd) analysis of Sentinel-2 data.

Details on the NVG “clear cuts” reference data set are provided in

a)

https://github.com/S2change/vegetation_loss/blob/main/data_info/reference_data/Harmonized/NVG/NVG.md and in

b)

https://github.com/S2change/vegetation_loss/blob/main/data_info/reference_data/NVG/readme.md.

2. Visual chips 4 km by 4 km (vchips). Those were created in areas around some of the features of the harmonized data set and were visually interpreted from Sentinel-2 imagery and other sources. This is described at https://github.com/S2change/vegetation_loss/blob/main/data_info/reference_data/vchips_for_training/%20readme.md.

When new maps will be produced and validated, this data set can be easily expanded, and used to fine-tune and improve the deep learning model.

Independent *reference data test validation results*

The results in the section refer to the deep learning models applied to the visual chips (vchips). Those validation results are complementary to an overall validation of the National Vegetation Loss Maps.

The “Visual Chips” data set was randomly partition in “train” (70%), “validation” (15%) and “test” (15%) data sets.

In this Section, detailed results are presented, both at the pixel level and at the patch level. A summary of results is available in the Conclusions.

EfficientNet_b2_16bit test results (180 test chips):

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PIXEL-LEVEL CLASSIFICATION REPORT (label patches < 25 px excluded)

Spatial mask applied: DGT_Mask_dissolve.gpkg

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precision	recall	f1-score	support
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Background	0.99	0.97	0.98	4578286
Cuts	0.71	0.57	0.63	44110
Fires	0.97	0.93	0.95	534790
micro avg	0.98	0.96	0.97	5157186
macro avg	0.89	0.82	0.85	5157186
weighted avg	0.98	0.96	0.97	5157186

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PATCH-LEVEL CONFUSION MATRIX

Rows = true label class | Cols = dominant predicted class
'missed' col = label patch has no non-bg prediction overlap

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	missed	Cuts	Fires	total	patch_recall
Cuts	23	46	0	69	0.67
Fires	8	0	58	66	0.88

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FALSE-ALARM PATCHES (prediction patches with no label overlap)

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Cuts	:	11 / 50 pred patches are false alarms (22.0%)
Fires	:	3 / 66 pred patches are false alarms (4.5%)

BACDM test results (58 chips):

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PIXEL-LEVEL CLASSIFICATION REPORT (label patches < 25 px excluded)

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	precision	recall	f1-score	support
Background	1.00	0.98	0.99	3417242
Cuts	0.52	0.69	0.59	27289
Fires	0.91	0.98	0.94	356432
accuracy			0.98	3800963
macro avg	0.81	0.88	0.84	3800963
weighted avg	0.98	0.98	0.98	3800963

=====

PATCH-LEVEL CONFUSION MATRIX

Rows = true label class | Cols = dominant predicted class

'missed' col = label patch has no non-bg prediction overlap

=====

	missed	Cuts	Fires	total	patch_recall
Cuts	18	50	0	68	0.74
Fires	0	0	42	42	1.00

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FALSE-ALARM PATCHES (prediction patches with no label overlap)

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Cuts : 47 / 98 pred patches are false alarms (48.0%)
Fires : 40 / 73 pred patches are false alarms (54.8%)

Concrete examples of predictions

In this section, some examples with independent test chips (2560 m by 2560 m) are illustrated. Below, we show a few examples, comparing reference data with predictions by EfficientNet_b2 and BACDM. The background is the “after” color composite for the analyzed date (which is available on the title).

vchip_664835_4427005_20190612_03

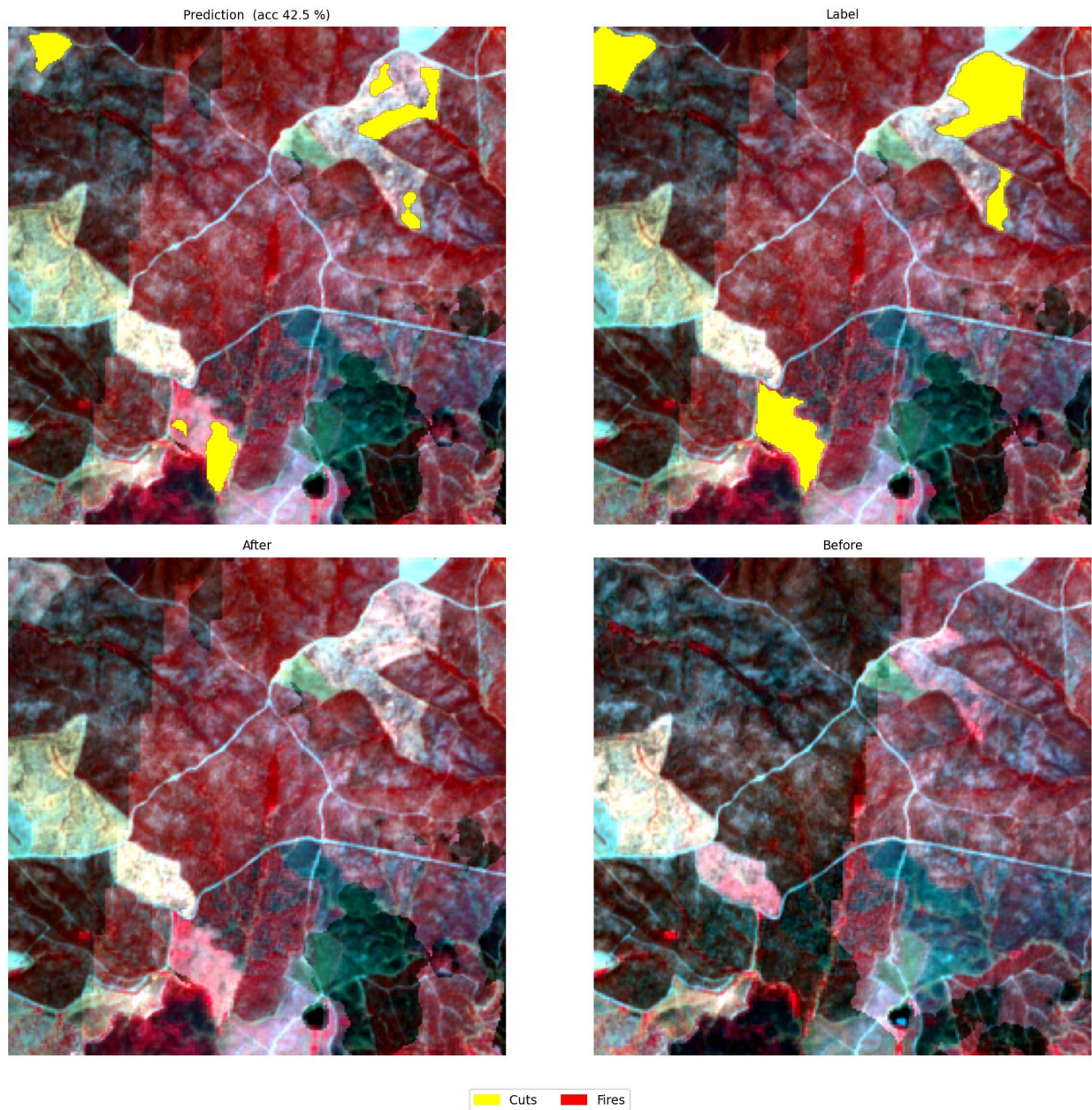


Figure 1: EffcientNet b2 prediction for a "Cut" patch. The lower images are respectively, "after" and "before" composites. The top images are respectively, the classifier prediction (yellow for cuts) and the reference map created by manual labeling from visualization. The title "vchip_date_x_y_" indicates the date of vegetation loss and the coordinates (x,y) of the centroid of the figure (CRS: EPSG 32629).

vchip_664835_4427005_20190612_03

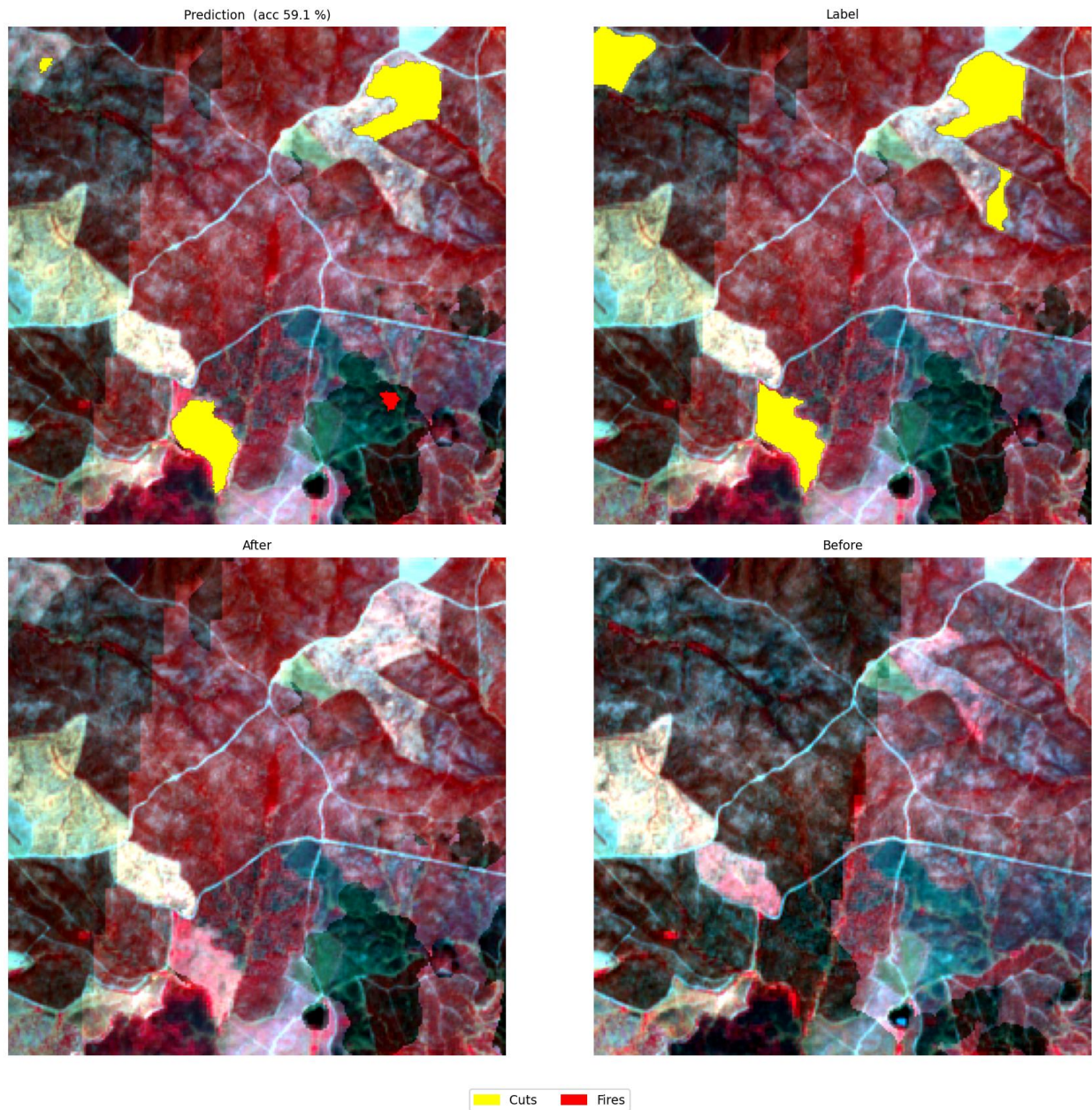


Figure 2: BACDM prediction for a "Cut" patch. The lower images are respectively, "after" and "before" composites. The top images are respectively, the classifier prediction (yellow for cuts) and the reference map created by manual labeling from visualization. The title "vchip_date_x_y_" indicates the date of vegetation loss and the coordinates (x,y) of the centroid of the figure (CRS: EPSG 32629).

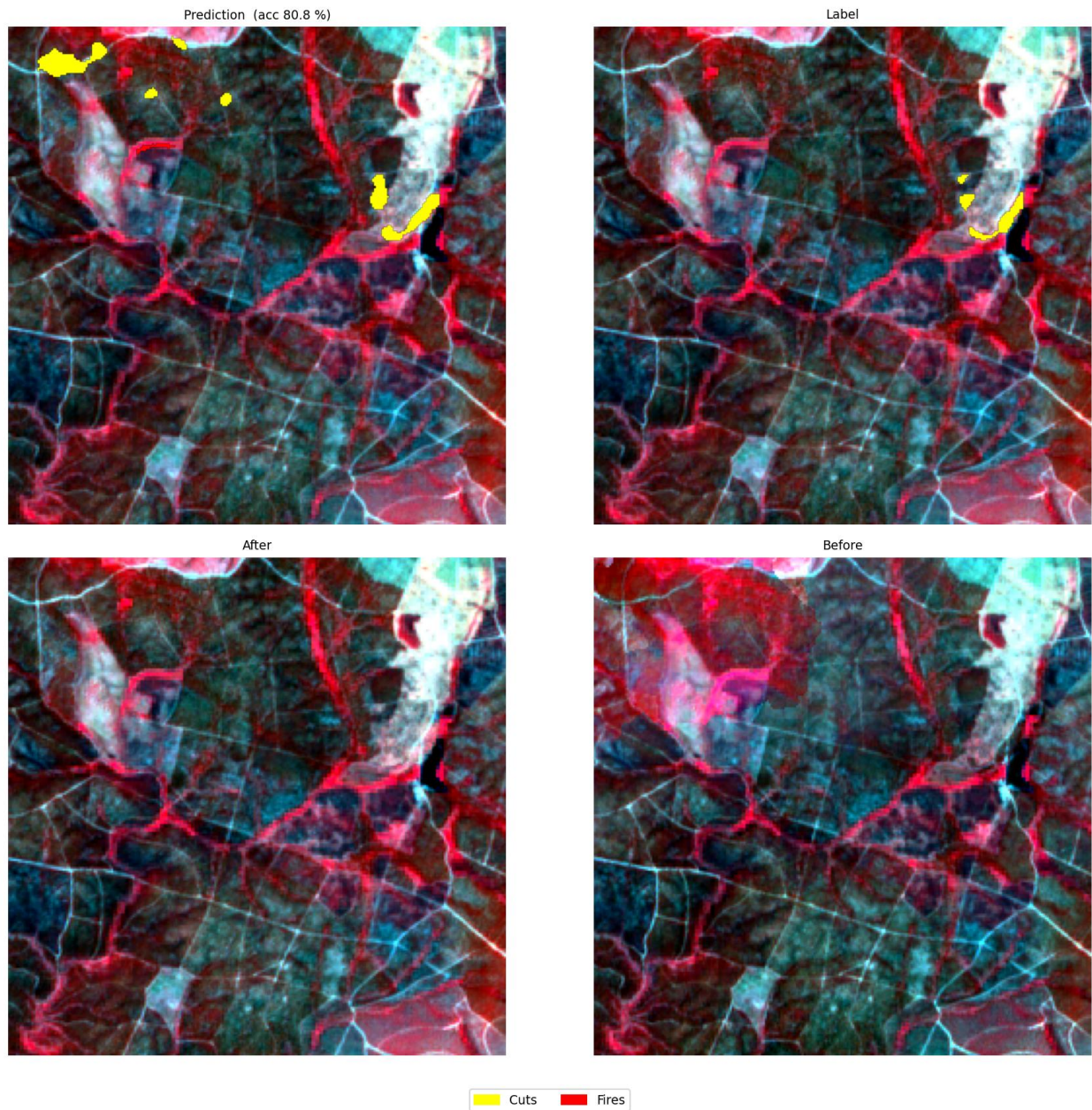


Figure 3: EffcientNet b2 prediction for a "Cut" patch. In this example, we can see the effect of cloud/haze on the "Before" image which caused commission in the prediction.

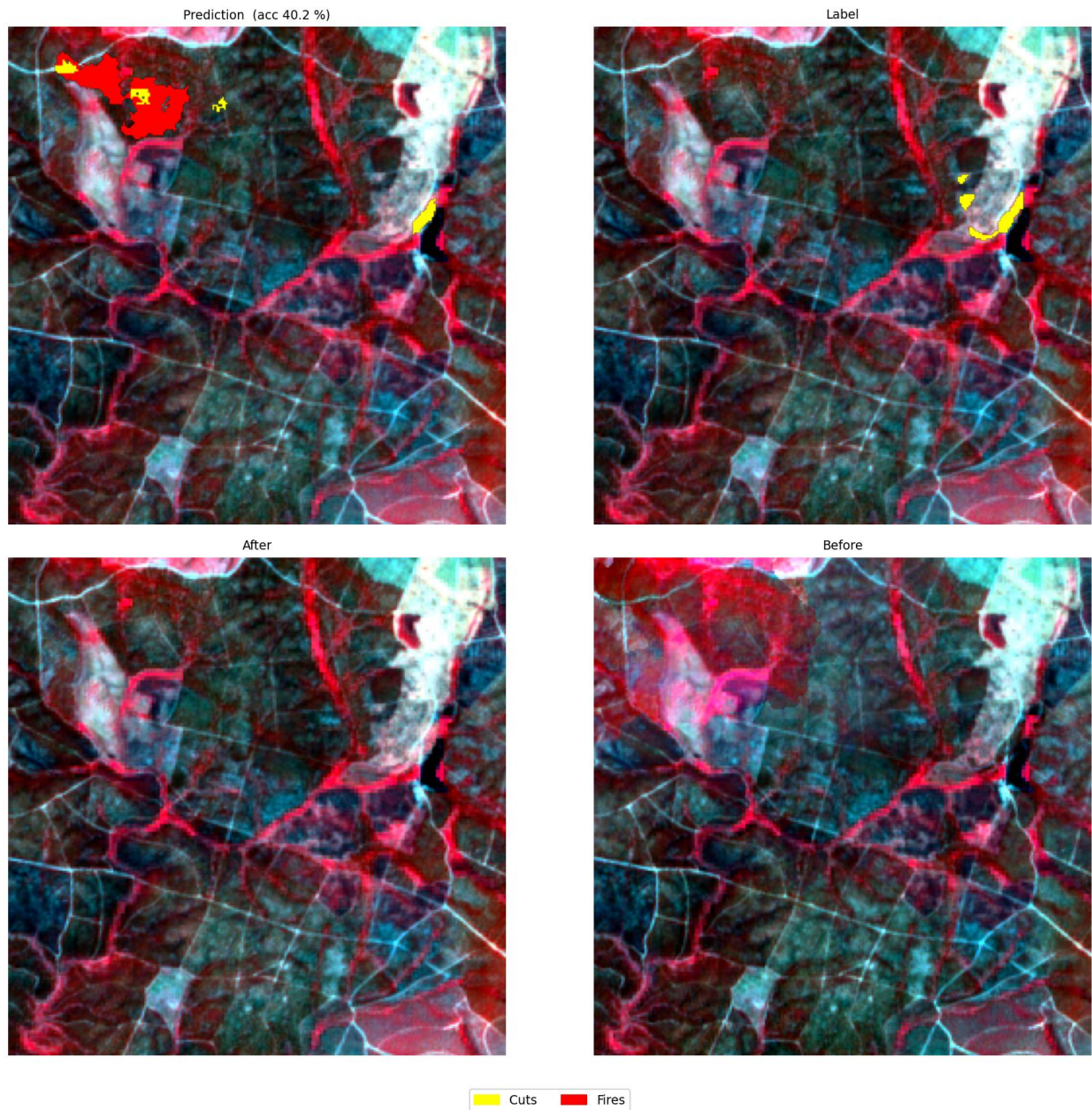


Figure 4: BACDM prediction for a "Cut" patch. In this example, we can see the effect of cloud/haze on the "Before" image which caused commission in the prediction and also confusion with the "Fires" class.

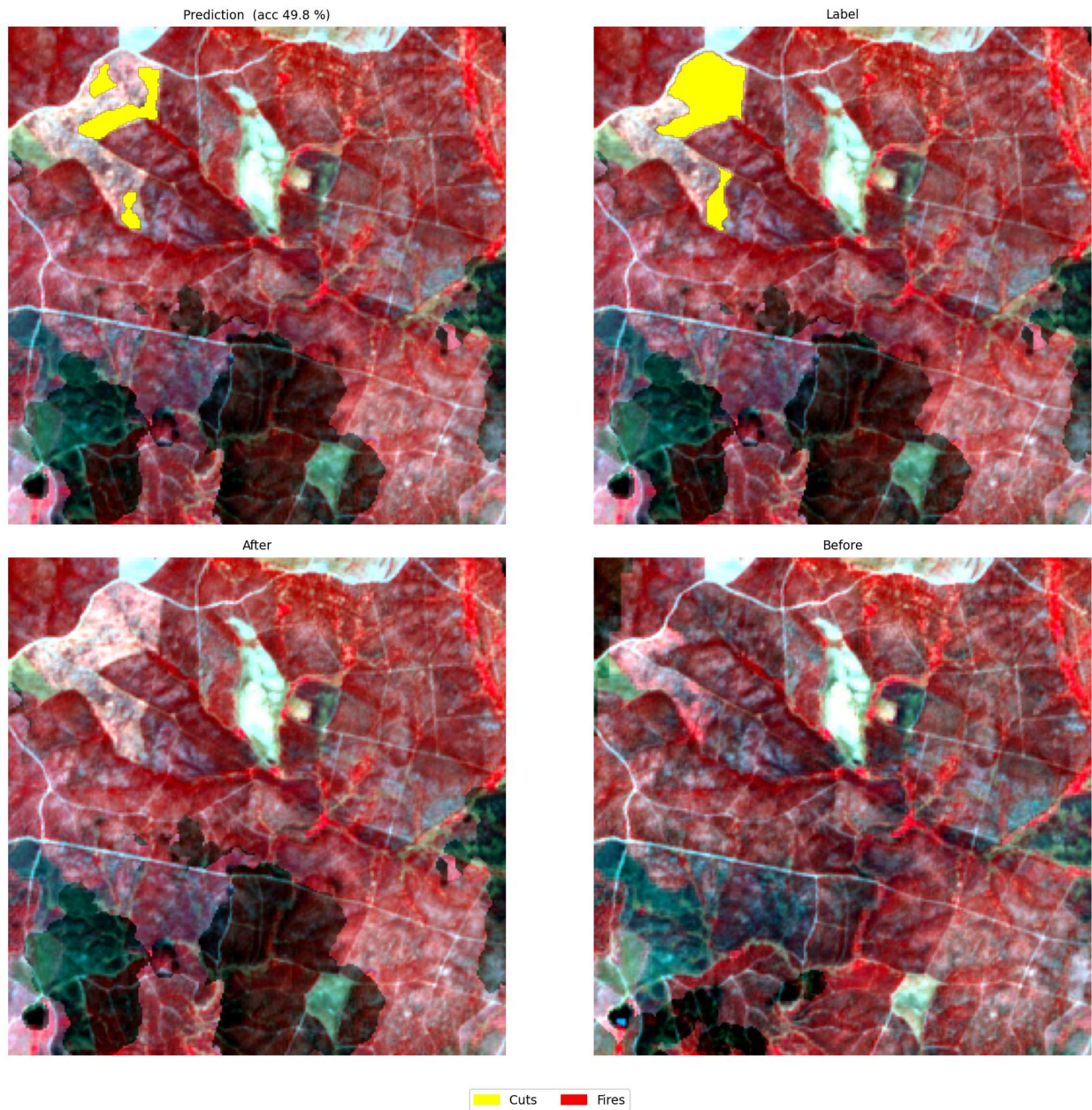


Figure 5: EffcientNet b2 prediction for a "Cut" patch. In this example, the predicted patches match the correct location but there is some omission in the prediction.

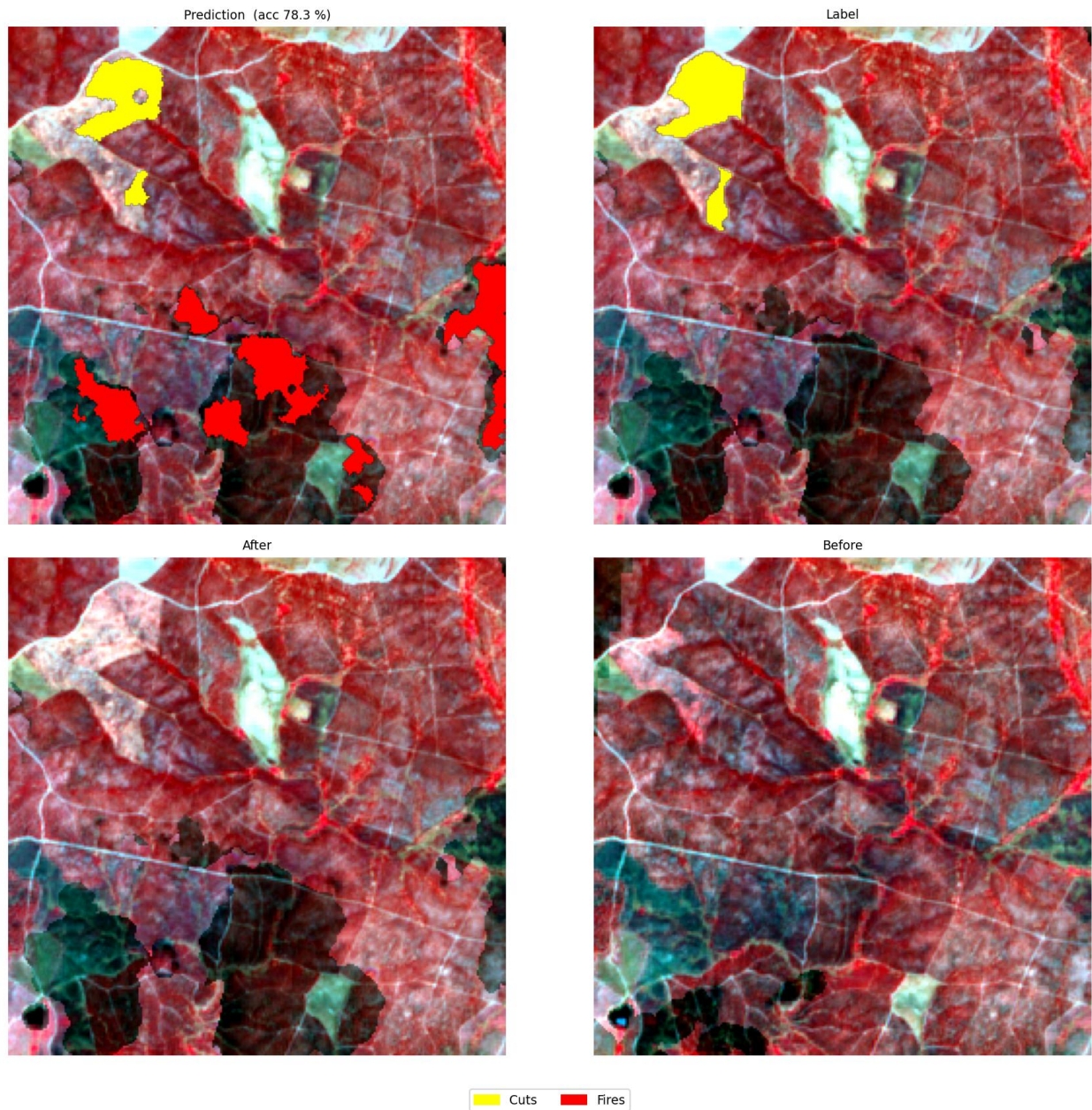


Figure 6: BACDM prediction for a "Cut" patch. In this example, the predicted "Cut" patches match the correct location but BCDM creates "Fires" patches where shadows contaminate the "after" time composite.

vchip_596775_4424835_20200913_03

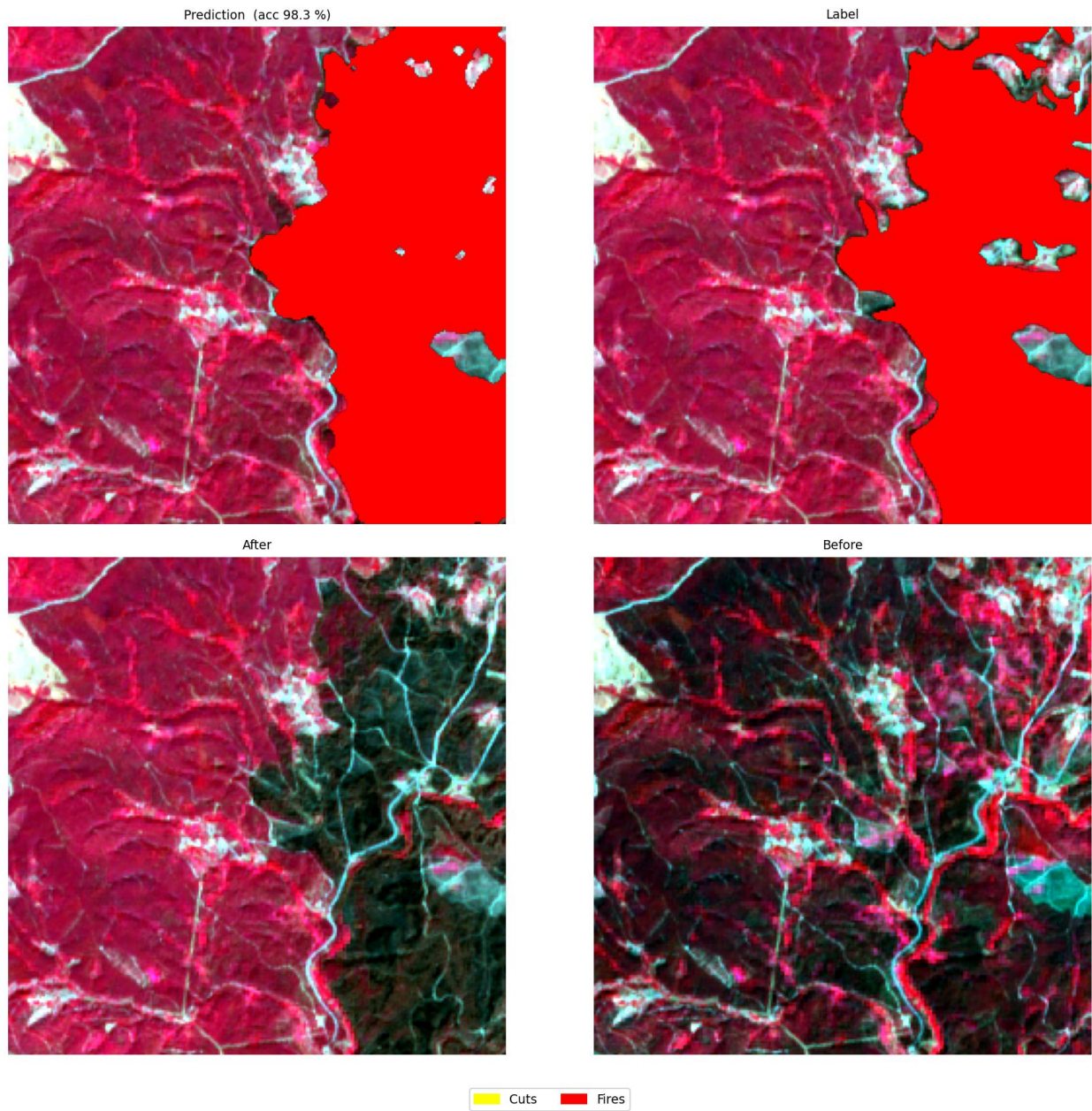


Figure 7: EffcientNet b2 prediction for a "Fire" patch.

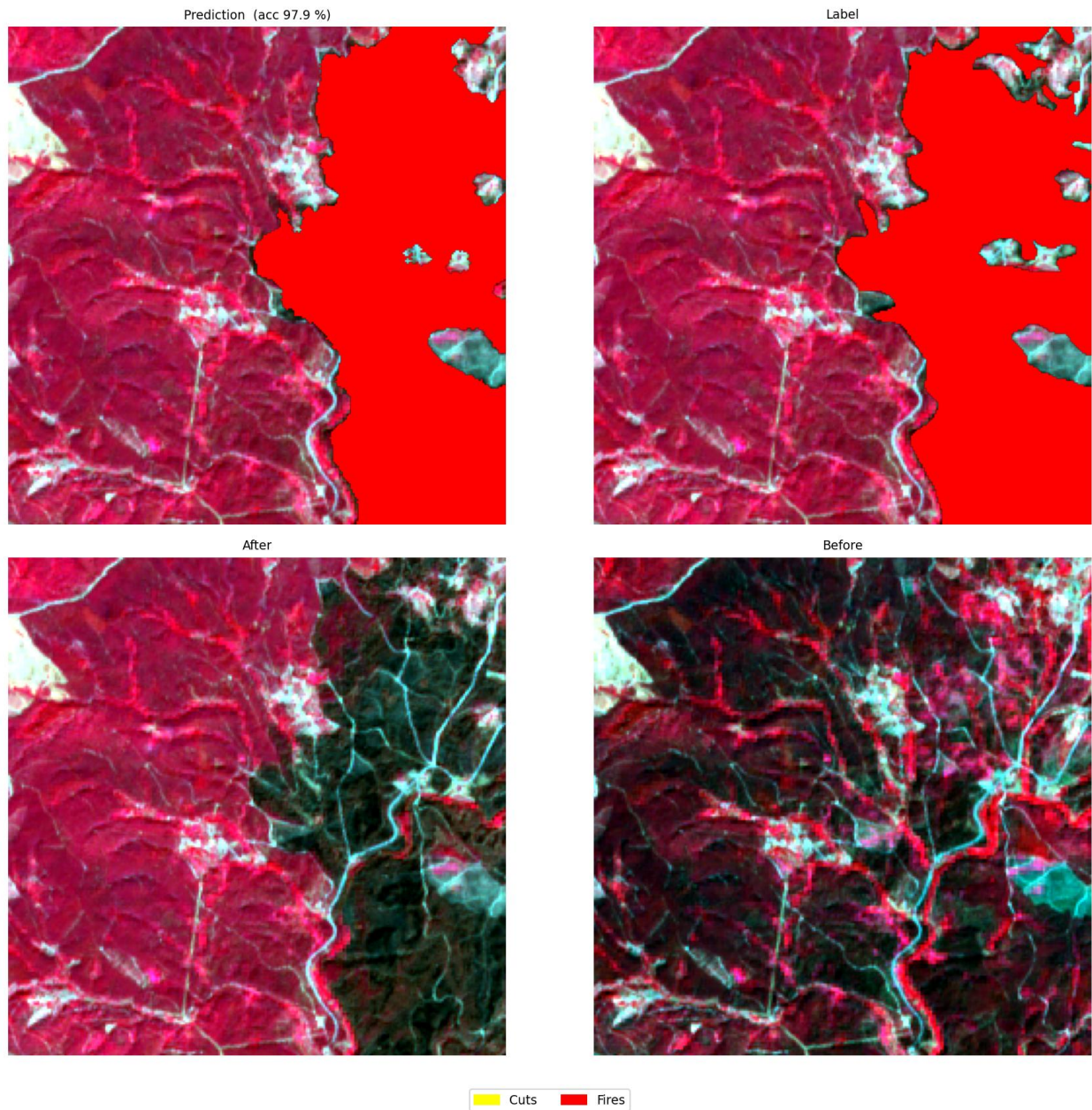


Figure 8: BACDM prediction for a "Fire" patch.

Some examples not in the reference data sets

The next examples show the Efficientnet_b2 predictions for areas a few areas of Portugal over a current Google Satellite image.

Figure 9 show an example of a "Cuts" prediction and Figure 10 show also an example of adjacent "Cuts" with distinct dates. Figure 11 illustrates the detection of "Fires".



Figure 9: Example of a (vector) feature with a detection of “Cuts” (label=1) at location Long= -8.90789, Lat=39.65848. The background Google Imagery is from 2025-2026.



Figure 10: Adjacent detections in tile T29TNE. The background Google Imagery is from 2025-2026.



Figure 11: Example of a (vector) feature with a detection of “Fires” (label=2) at location Long= -7.61088 , Lat=40.23848. The background Google Imagery is from 2025-2026.

On the areas where tiles overlap, the range of dates is larger, and the same change event can be identified at distinct dates, as illustrated in Figure 12.

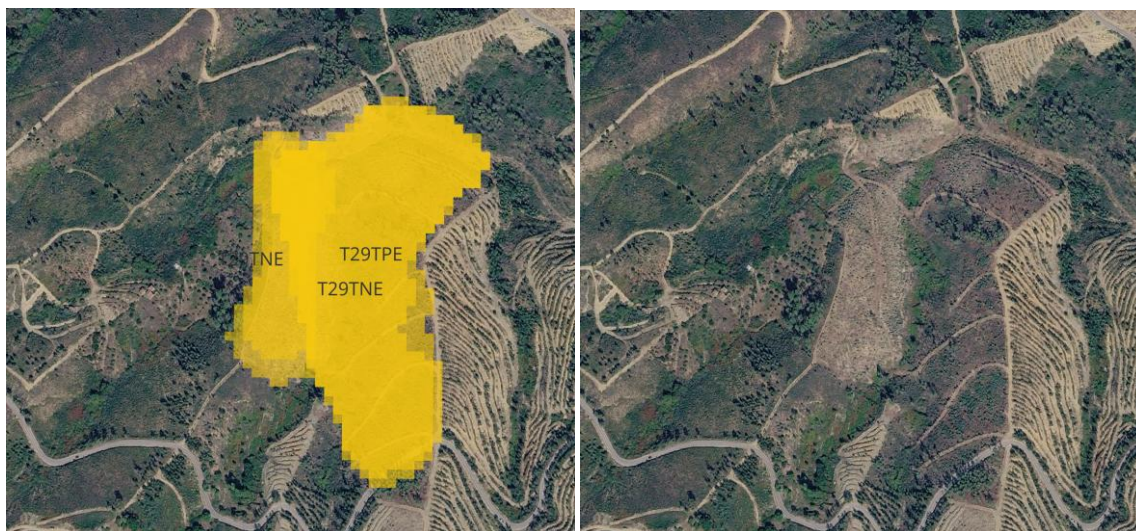


Figure 12: Detections of “Cuts” on the intersection of tiles T29TNE and T29TPE. The background Google Imagery is from 2025-2026 (Long= -7.79619, Lat=39.96118)

For the remainder “Fires” predictions, we compare visually the prediction with the official burned area map from ICNF.



Figure 13. Prediction around location long=-7.10888,lat=40.15102 (near Penamacor) . The predicted Cuts (in yellow) in general match the eucalyptus stands in the area. One can see some small patches that must be commissions.

Results for the area where large fires occurred near Ourém in the summer of 2023 show a very good match between the ICNF burned area maps and the predictions from the ISA_DGT_INCD_DL_pipeline (see Figure 14 and Figure 15).

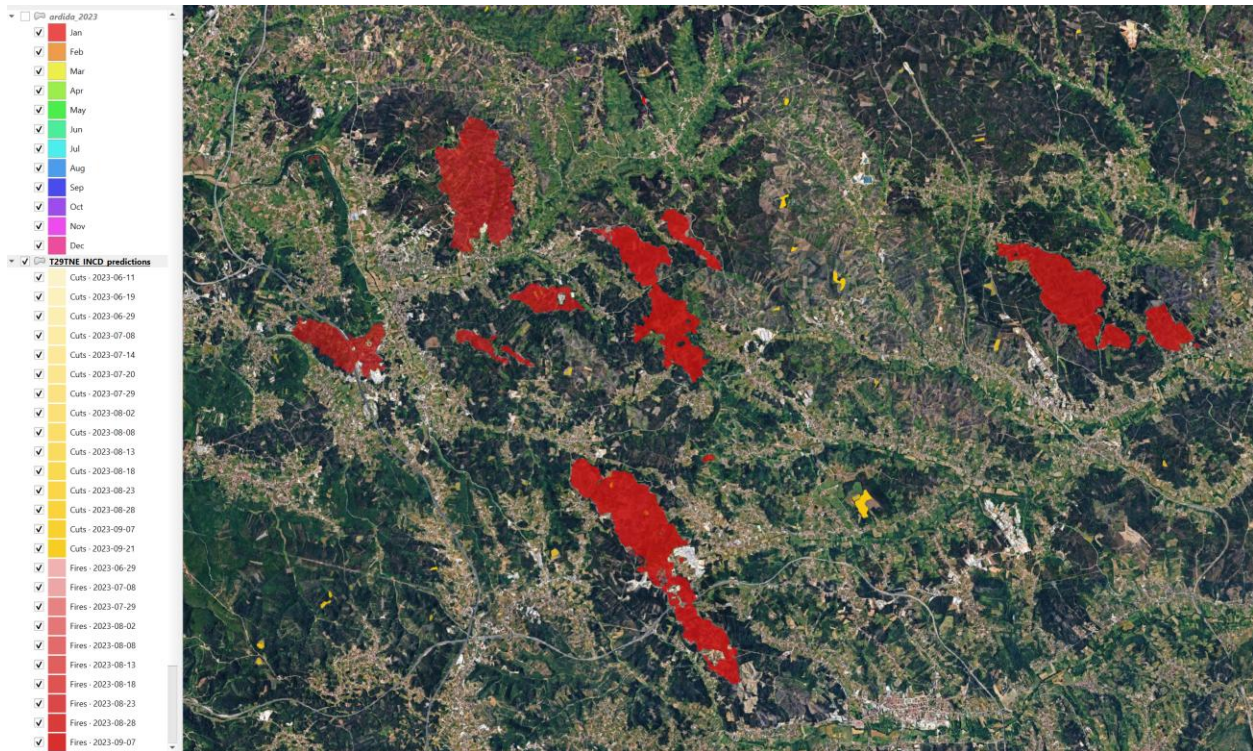


Figure 14: Detections returned by ISA_DGT_INCD_DL_pipeline for the region of Ourém in 2023 (entered at Long=-8.60238,Lat=39.71876 and approximately 22 km E-W and 16 km N-S)

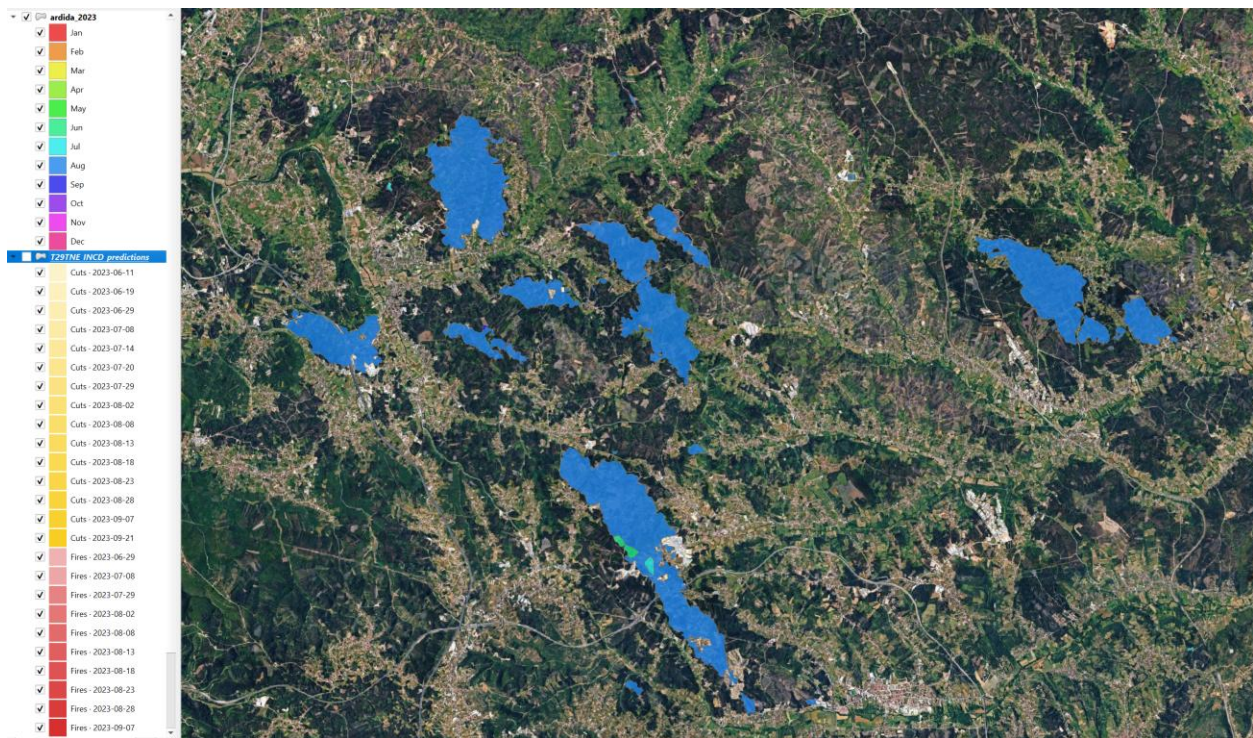


Figure 15: Burned areas from the official ICNF map for 2023.

Conclusions

Validation at the chip level for Efficientnet_b2 shows that mapping has high precision for “Fires” (precision= 0.97; recall=0.93; F1-score=95%) and moderate for “Cuts” (precision= 0.71; recall=0.57; F1-score=63%) at the pixel level.

For patches, results for independent test chips suggest that 8% of Fires are missed, and 3% of Fires correspond to False alarms. For Cuts, the estimates are 23% of Cuts missed and 22% are False alarms.

The availability of an extensive reference data set will allow to extend the validation results to areas outside the reference visual chips used in this report.

The reference data set can be easily extended. Namely, the set of visual chips can be enriched to allow further fine tuning of the deep learning model.

Those changes can be easily incorporated in the current ISA_DGT_INCD_DL_pipeline.